ITE4116M FINAL YEAR PROJECT

PROJECT SUGGESTION FORM

Academic Year 2026/27

Submission Guide: Students must form a group of 3 to 4 members, discuss the project topic with your preferred supervisor, and obtain verbal agreement. Once agreed, the group must complete this Project Suggestion Document and upload the document file (**in both PDF & DOCX**) on Moodle during the FYP session by **14-SEP-2026.**

1. General Project Information

Proposed Project Title	AI and IoT Based Food Replenishment and Waste Reduction System
Assigned Project Supervisor	Macro (proposed; verbal agreement pending)
Programme / Class Code	Higher Diploma in AI and Smart Technology (IT114124) / Class AIST-2B
Student Group Number	Group 2

Student Member Details (Normally 3 to 4 members):

S/N	Student Name	Student ID	Email Address	Proposed Major Role / Area of Responsibility
1	Yuan Yuchen	250342840	250342840@stu.vtc.edu.hk	**Project Leader and Backend AI Lead** **Planned workload: 40%** Lead requirements and architecture. Develop APIs, food recognition, demand forecasting and replenishment logic. Coordinate integration, evaluation and project documentation.

2	Xu Ziqian	250137555	250137555@stu.vtc.edu.hk	**IoT Hardware and Edge Processing Developer** **Planned workload: 30%** Build sensing hardware and Raspberry Pi acquisition services. Implement calibration, filtering, tray events and recovery. Test measurement reliability and document installation.
3	Li Wenhua	250329909	250329909@stu.vtc.edu.hk	**Web Application and User Experience Developer** **Planned workload: 30%** Develop staff tools and QR menu/order pages. Implement live status, bilingual interfaces and API integration. Test usability and prepare user documentation.

2. Problem Statement & Project Objectives

Instructions: Describe the business or technical problem in information technology that this project intends to address. Define 2-3 specific project objectives (what your solution intends to achieve for the target users/businesses).

Problem Statement:

Buffet and batch-prepared self-service outlets must replenish food before trays run empty while avoiding excess food at closing. Visual checks and separate kitchen records can leave staff without timely information about remaining food, demand and batches already being prepared. Late or duplicate replenishment can cause waiting and avoidable leftovers, while customer menus may show outdated availability.

We propose an affordable prototype that continuously monitors trays while they remain in service and combines measurements with demand estimates and preparation progress. Metafoodx and Controliza are commercial references; our evaluation will compare the proposed workflow with simple replenishment baselines within a small serving line.

Project Objectives:

1. Design and implement continuous monitoring for up to three serving positions, recording remaining food, replenishment and surface temperature, with detection of tray removal and unreliable readings.
2. Develop AI-assisted demand estimates and replenishment recommendations using stock, preparation lead time and work in progress. Compare empty-tray time and closing leftovers against fixed-threshold and consumption-rate baselines.
3. Integrate bilingual staff tools and a supplementary QR menu/order interface. Evaluate data consistency, interruption recovery and usability through hardware experiments and controlled service scenarios.

3. Proposed Project Scope & Key Features

Instructions: Define the scope of your system. List and briefly describe the key modules, functions, or features that your group plans to implement. Projects must be implementation-oriented and student-centered.

Key Subsystems / Features:

A. Continuous tray monitoring. Trays remain on sensing bases throughout service. Combine weight, visual dish identification and surface-temperature readings; handle tare, movement, refills, occlusion and stale data.

B. Replenishment and kitchen tasks. Recommend start times and batch quantities using demand, preparation time, closing time and batches in progress. Staff accept, modify or reject recommendations; completed refills update the system.

C. Operations dashboard and evaluation. Show tray status and alerts; record empty-tray time, refill history, leftovers and disposal/reuse destinations. Export records for baseline comparison.

D. Supplementary QR menu and ordering. Show dish details, availability, completed-order popularity or buffet take-up trends, and estimated waits. Provide basic ordering and status for self-service fast casual and buffet made-to-order items; reconcile pending orders with physical stock.

Prototype Scope:
Six months, one serving line, up to three monitored positions and 4-6 registered dishes. Excludes payment processing, delivery, procurement ERP and independent food-safety certification.

Expected Users & Roles:
Kitchen/service staff monitor trays and manage refills. Managers maintain menus, settings and reports. Customers view the QR menu and submit or track applicable orders.

4. Technical Stack & Special Requirements

Instructions: Detail the hardware and software tools you intend to use. Highlight how your project incorporates contemporary technologies (AI, IoT, Cloud, etc.) and how it aligns with modern considerations like environmental-friendly design, bilingual/multi-language support, or cross-border deployability to meet Greater China business needs.

Development Technologies & Tools:
- **Hardware:** Raspberry Pi 5 with load cells and ADC interfaces (such as HX711 or NAU7802), an overhead camera, surface-temperature sensor, display and LEDs. Assess RGB-D after range and cost tests. Existing UNO R4 WiFi is optional for acquisition/debugging.
- **Programming Languages:** Python, TypeScript, HTML/CSS; C++ only if a microcontroller is used.
- **Frameworks / Libraries:** FastAPI, OpenCV, PyTorch for model training, scikit-learn, React and i18next.
- **Database / Storage:** PostgreSQL and local measurement/image logs.
- **Target Platforms:** Local Linux device and responsive staff/customer web interfaces.

Special Project Considerations:
- **Contemporary Technologies:** IoT sensing, local AI inference and live web updates; optional cloud backup after the local core works.
- **Environmental-Friendly Solution:** Evaluate avoidable food surplus; reuse available computing equipment and use replaceable sensing components.
- **Bilingual / Multi-Language Support:** English and Traditional Chinese interfaces.
- **Cross-Border Deployment / Greater China Needs:** Configurable language, currency and time zone, with local deployment options for later regional adaptation.
- **Feasibility:** Confirm camera/ADC compatibility, sampling stability and cost by bench tests. Use controlled service scenarios initially and arrange a field trial if access becomes available.

5. Endorsements & Supervisor Feedback

Student Group Endorsement

By submitting this suggestion, we confirm that all listed group members are in agreement regarding this project concept and our assigned roles.

Student Representative Signature: _____

Date: _____

Supervisor Vetting Decision

Supervisor, please put a ☑ in the appropriate box.

☐ **APPROVED — The group is authorized to proceed to draft their formal Project Proposal.**

☐ **REJECTED — The group must modify their project topic suggestion and resubmit for review.**

Supervisor Feedback / Comments

Supervisor Endorsement

Project Supervisor Signature: _____

Date: _____